
Deterministic Mathematical Modeling and Sizing Framework for Distributed Clean Energy

Solar Photovoltaics, Stationary Battery Storage, and Level 2 EV Infrastructure

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Scope. This paper presents deterministic, formula-backed sizing methods for distributed clean energy systems, with emphasis on transparent assumptions, reproducible calculations, and engineering-standard constraints across photovoltaic generation, battery storage, and Level 2 electric-vehicle charging infrastructure.

Abstract

Distributed energy resources (DERs)—including photovoltaic (PV) generation, battery energy storage systems (BESS), and electric vehicle supply equipment (EVSE)—form the cornerstone of modern residential and commercial decarbonization. However, educational institutions and engineering practitioners frequently encounter sizing calculators obscured by vendor bias, concealed safety margins, or inaccurate linear approximations. This paper provides a rigorous, formula-backed mathematical framework for sizing distributed clean energy systems. We detail deterministic formulations for sub-zero PV open-circuit voltage expansion (V_{oc}), electrochemical depth-of-discharge and Peukert capacity derating, solar angle of incidence capture, and continuous-load circuit sizing under IEEE, IEC, and NFPA 70 (NEC) standards.

Keywords: distributed energy resources; solar photovoltaic; battery energy storage; EVSE; battery runtime; Peukert’s law; cold-temperature voltage; solar geometry; continuous-load sizing; NEC.

1. Introduction & Methodology

The transition toward decentralized electrification requires multi-domain energy balance across three interacting subsystems:

$$\text{Generation (Solar PV)} \longleftrightarrow \text{Storage (BESS)} \longleftrightarrow \text{High-Power Loads (EV \& HVAC)} \quad (1)$$

Many public calculation tools omit conversion losses, temperature coefficients, and non-linear discharge kinetics. The objective of this open framework is to establish transparent mathematical parity between electrical theory and real-world system design.

2. Electrochemical Energy Storage & Discharge Kinetics

2.1 Nominal vs. Delivered Usable Energy

Battery capacity rated in Amp-hours (Ah) is physically incomplete without operating nominal potential (V_{nom}). The nominal energy storage capacity E_{nom} is defined as:

$$E_{nom} = C_{nom} \times V_{nom} \quad (\text{Watt-hours, Wh}) \quad (2)$$

In practical stationary and off-grid applications, delivered AC energy $E_{delivered}$ is constrained by usable Depth of Discharge (DoD_{usable}), Battery State of Health (SoH), and power conversion efficiency (η_{inv}):

$$E_{delivered} = E_{nom} \times DoD_{usable} \times SoH \times \eta_{inv} \quad (3)$$

2.2 Inverter Tare Loss and Total Load Power

Power inverters consume continuous standby power (P_{tare} , typically 10W–35W) independent of output load. For an AC appliance load P_{load} operating with duty cycle D , the effective continuous battery-side power demand P_{bat_side} is:

$$P_{bat_side} = \left(\frac{P_{load} \times D}{\eta_{inv}} \right) + P_{tare} \quad (4)$$

The continuous backup runtime duration $T_{runtime}$ (in hours) is expressed as:

$$T_{runtime} = \frac{C_{nom} \times V_{nom} \times DoD_{usable} \times SoH}{\left(\frac{P_{load} \times D}{\eta_{inv}} \right) + P_{tare}} \quad (5)$$

2.3 Non-Linear Rate Capacity Derating (Peukert's Law)

For lead-acid, AGM, and gel chemistries, effective capacity diminishes under high discharge rates (C -rate) due to electrolyte diffusion limitations (Wilhelm Peukert, 1897):

$$C_{effective} = C_{rated} \left(\frac{I_{rated}}{I_{actual}} \right)^{k-1} \quad (6)$$

Where:

- C_{rated} = Manufacturer nominal capacity at rated 20-hour rate ($I_{rated} = C_{rated}/20$).
- I_{actual} = Real-time continuous discharge amperage (P_{bat_side}/V_{bat}).
- k = Empirical Peukert exponent (LiFePO₄: $1.02 \leq k \leq 1.05$; AGM: $1.12 \leq k \leq 1.20$; Flooded Lead-Acid: $1.20 \leq k \leq 1.35$).

3. Photovoltaic Thermal-Voltage & Geometric Irradiance Modeling

3.1 Cold-Temperature Open-Circuit Voltage (V_{oc}) Expansion

Photovoltaic silicon exhibits a negative temperature coefficient of voltage ($\gamma_{V_{oc}}$, typically -0.28% to $-0.35\%/^{\circ}\text{C}$). As ambient temperature decreases below Standard Test Conditions ($T_{STC} = 25^{\circ}\text{C}$), array open-circuit voltage rises substantially.

Under NFPA 70 / NEC Article 690.7(A), the maximum cold-temperature string voltage V_{oc_cold} must not exceed the charge controller or inverter maximum input voltage (V_{input_max}):

$$V_{oc_cold} = V_{oc_STC} \times \left[1 + \left(\frac{|\gamma_{V_{oc}}|}{100} \right) \times (25^{\circ}\text{C} - T_{min}) \right] \times N_{series} \quad (7)$$

Where:

- T_{min} = Site historical record minimum winter ambient temperature ($^{\circ}\text{C}$).
- N_{series} = Number of PV modules wired in series per string.

3.2 Geometric Irradiance Capture & Cosine Incidence Angle

The effective solar flux $E_{effective}$ incident on a collector surface inclined at tilt angle β and azimuth γ_{panel} is governed by the angle of incidence θ :

$$E_{effective} = E_{DNI} \cos(\theta) + E_{diffuse} \quad (8)$$

$$\cos(\theta) = \sin(\alpha) \cos(\beta) + \cos(\alpha) \sin(\beta) \cos(\gamma_{sun} - \gamma_{panel}) \quad (9)$$

Where α is the solar altitude angle and γ_{sun} is the solar azimuth position.

4. Continuous-Load Electrical Infrastructure & EVSE Sizing

4.1 NEC 125% Continuous Duty Mandate (The 80% Rule)

Under NFPA 70 / NEC Article 625.41, electric vehicle charging loads operate continuously for ≥ 3 consecutive hours. To prevent thermal cumulative heating inside overcurrent protection devices (OCPD),

circuits must be oversized by 125%:

$$I_{breaker} \geq I_{continuous} \times 1.25 \quad \Longleftrightarrow \quad I_{continuous} \leq I_{breaker} \times 0.80 \quad (10)$$

Table 1: Standard Conductor & Breaker Sizing for Level 2 EVSE (240V Split-Phase)

Charger Continuous Current	Required OCPD Breaker	Conductor Gauge (THHN Copper @ 75°C)	Delivered Power (kW)	Range Replenishment Rate
16 Amps	20 Amps	12 AWG	3.84 kW	12–15 miles/hr
32 Amps	40 Amps	8 AWG	7.68 kW	25–32 miles/hr
40 Amps (NEMA 14-50 Max)	50 Amps	6 AWG	9.60 kW	30–38 miles/hr
48 Amps (Hardwire Only)	60 Amps	6 AWG (4 AWG NM-B)	11.52 kW	36–46 miles/hr
80 Amps (High-Power Fleet)	100 Amps	3 AWG / 2 AWG	19.20 kW	60–75 miles/hr

5. Implementation & Open Access Distribution

The mathematical formulations described in this paper are fully implemented in an open-access, zero-database software platform built with deterministic TypeScript calculation engines:

- **Platform Root:** <https://www.powelab.org>
- **Battery Sizing Engine:** powelab.org/guides/battery-backup-runtime-calculation-guide
- **Solar PV Modeling:** powelab.org/guides/solar-panel-tilt-angle-by-latitude-and-season-guide
- **EV Charging Sizing:** powelab.org/guides/level-2-ev-charging-speed-and-breaker-sizing-guide

6. References & Engineering Standards

1. **IEEE Std 485-2020:** *IEEE Recommended Practice for Sizing Lead-Acid Batteries for Stationary Applications.*
2. **NFPA 70 / National Electrical Code (NEC) 2023:** *Articles 220 (Branch Circuit Calculations), 625 (Electric Vehicle Power Transfer Systems), 690 (Solar Photovoltaic Systems), and 706 (Energy Storage Systems).*
3. **National Renewable Energy Laboratory (NREL):** *PVWatts Version 8 Technical Reference Manual & Algorithms.*
4. **IEC 62619:** *Secondary Cells and Batteries for Use in Industrial Applications.*
5. **IEC 62548:** *Photovoltaic (PV) Arrays - Design Requirements.*
6. **SAE International:** *SAE J1772 & SAE J3400 Electric Vehicle Conductive Charge Coupler Standards.*
7. **Peukert, W. (1897):** *Über die Abhängigkeit der Kapazität der Bleiakumulatoren von der Entladestromstärke.* Zeitschrift für Elektrotechnik.

Educational and engineering-use notice. This open educational paper provides deterministic calculation methods and is not a substitute for project-specific engineering review, equipment manufacturer instructions, applicable local electrical codes, conductor derating calculations, permitting requirements, or inspection by the authority having jurisdiction (AHJ). Field installations should be designed and verified by appropriately qualified professionals.